InternetSim CLI

Full Session Documentation

BGP Routing Simulation • Policy Modification • ROV Analysis

1. Session Overview

This document records a complete InternetSim CLI session demonstrating BGP prefix announcement, route querying, routing policy modification, ROV (Route Origin Validation) configuration, and topology reloading. The session was executed against a real AS topology of 76,802 ASes and 495,579 links.

2. Environment Setup

2.1 Loading Configuration Files

Two files were loaded before initialization:

```
load ASRelFilePath = data/20240501.as-rel.txt;
load InvalidSetFilePath = data/Invalid-Set.txt;
```

Key	File Path	Purpose
`ASRelFilePath`	`data/20240501.as-rel.txt`	AS relationship topology (peers, providers, customers)
`InvalidSetFilePath`	`data/Invalid-Set.txt`	Set of invalid/hijacked prefixes

2.2 Engine Initialization

The engine was initialized with init;, loading the full topology:

```
init;
```

Output:

```
Loaded 76802 ASes, 495579 links
Loaded 1 Invalid-SET
  Engine initialized
```

Metric	Value
Total ASes loaded	76,802
Total links loaded	495,579
Invalid-SET entries	1

3. BGP Prefix Announcements

3.1 Initial Announcements

Eight ASes announced prefixes spanning multiple IP ranges:

```
AS 174  announce 187.156.112.0/23;
AS 209  announce 149.181.128.0/20;
AS 286  announce 29.214.0.0/16;
AS 1239 announce 186.80.0.0/13;
AS 1299 announce 128.126.128.0/18;
AS 2828 announce 86.194.0.0/15;
AS 2914 announce 108.0.0.0/8;
AS 3257 announce 10.0.0.0/8;
Simulate;
```

3.2 Simulation Results — Round 1

Metric	Value
Convergence time	5.6546 seconds
BGP updates	1,366,320
Affected ASes	76,228

4. Route Query — Before Policy Change

Routes to prefix 29.214.0.0/16 at AS 34549 before any policy was applied:

```
AS 34549 show routes 29.214.0.0/16;
```

ASPath	localPref	Origin
286	80	IGP
1299 286	60	IGP
174 286	60	IGP
3257 286	60	IGP
3356 286	60	IGP
3320 286	60	IGP
5511 286	60	IGP
6830 286	60	IGP
3223 1299 286	60	IGP

Observation: The direct route via AS 286 holds the highest local preference (80). All other paths arrive via transit ASes with localPref 60.

5. Routing Policy Modification

5.1 Policy Definition

A filter-in rule was added at AS 34549 for traffic arriving from peer AS 3356, elevating local-pref for prefix 29.214.0.0/16 to 120:

```
Change routing policy:
AS 34549 peer AS 3356
    filter in
        show rule
        add rule
            match "prefix is 29.214.0.0/16"
            action "local-pref 120"
        end
        show rule
    end
end;
```

5.2 Policy Engine Output

```
AS 34549 peer AS 3356 filter in no rules
AS 34549 peer AS 3356 filter in rule 0: match "prefix is 29.214.0.0/16", action
"local-pref 120"
Simulation complete
Convergence time: 0.001209 seconds
BGP updates: 73
Affected ASes: 60
```

The incremental simulation was extremely fast (1.2 ms) since the policy change only affected 60 ASes.

5.3 Route Query After Policy Change

The path via AS 3356 now holds the highest local preference (120), displacing the direct route:

ASPath	localPref	Notes
3356 286	120	Elevated by policy rule 0
286	80	Direct route — still best without policy
1299 286	60	
174 286	60	
3257 286	60	
3320 286	60	
5511 286	60	
6830 286	60	
3223 1299 286	60	

6. BGP Hijack Scenario

6.1 Competing Announcements

Two additional ASes announced the same prefix 187.156.112.139/23, simulating a BGP prefix hijack:

```
AS 52866 announce 187.156.112.139/23;
Simulate;
AS 8309  announce 187.156.112.139/23;  // Hijacker mentioned in InvalidSet file
Simulate;
```

Round	Announcer	Convergence Time	BGP Updates	Affected ASes
1	AS 52866	0.9888 s	529,289	76,382
2	AS 8309	0.8880 s	341,452	72,435

```
AS 28223 show routes 187.156.112.139/23;
```

6.2 Route Table at AS 28223 (Pre-ROV)

AS 28223 received routes from both hijacking ASes with mixed local preferences:

ASPath (excerpt)	localPref	Terminal AS
... 20473 52025 8309	80	8309 (hijacker)
... 52025 8309	80	8309 (hijacker)
47787 6424 8309	80	8309 (hijacker)
61568 14840 1031 40850 52866	80	52866
6057 262459 262907 52866	80	52866
271253 40850 52866	80	52866
37468 ... 8309	60	8309 (hijacker)
28158 6939 8309	60	8309 (hijacker)
264409 52866	60	52866

7. Non-BGP Policy — ROV Enablement

7.1 Policy Definition

ROV was enabled at four transit ASes to filter invalid route announcements:

```
Change non-BGP policy:
AS 20473 enable ROV
AS 52025 enable ROV
AS 6939  enable ROV
AS 6424  enable ROV
end;
```

Metric	Value
Convergence time	1.6708 seconds
BGP updates	767,772
Affected ASes	71,589

7.2 Route Table at AS 28223 (Post-ROV)

After ROV enforcement, routes transiting the four ROV-enabled ASes were suppressed. The table shrank significantly:

ASPath (excerpt)	localPref	Observation
47787 6424 1031 40850 52866	80	now ROV-enabled; path persists via 52866
61568 14840 1031 40850 52866	80	Path via 52866
6057 262459 262907 52866	80	Path via 52866
271253 40850 52866	80	Path via 52866
28158 271253 40850 52866	60	
37468 28220 1031 40850 52866	60	
264409 52866	60	

Routes through AS 20473 and AS 52025 (which were heavily used by AS 8309) were eliminated, substantially reducing that hijacker's reach.

8. ROV Topology Expansion

8.1 Loading High-Degree ROV Set

```
load ROVSetFilePath = data/rov_highdeg_10.txt;
reload;
```

```
Loaded 76802 ASes, 991158 links
Loaded 7680 ROV-SET
Loaded 1 Invalid-SET
  Reloaded (state preserved)
```

Metric	Before reload	After reload	Delta
Total links	495,579	991,158	+495,579 (+100%)
ROV-enabled ASes	4 (manual)	7,680	+7,676
Invalid-SET	1	1	unchanged

8.2 Route Query at AS 28158 (After reload, Before re-init)

ASPath (excerpt)	localPref
47787 6424 1031 40850 52866	80
6057 262459 262907 52866	80
271253 40850 52866	80
37468 28220 1031 40850 52866	60
28158 271253 40850 52866	60

9. Full Re-initialization and Final Simulation

9.1 Re-init with ROV Set

The engine was re-initialized to apply both the expanded link graph and the 7,680-AS ROV set from a clean state:

```
init;
AS 52866 announce 187.156.112.139/23;
AS 8309  announce 187.156.112.139/23;
Simulate;
```

Metric	Value
ASes loaded	76,802
Links loaded	495,579
ROV-SET entries	7,680
Convergence time	1.000077 seconds
BGP updates	530,088
Affected ASes	76,382

9.2 Final Route Table at AS 28223

With the high-degree ROV set active, paths through many major transit providers were filtered. The route table now only surfaces paths through unfiltered ASes:

ASPath (excerpt)	localPref	Terminal
35280 7545 ... 262907 52866	80	52866
61568 14840 1031 40850 52866	80	52866
47787 6424 1031 40850 52866	80	52866
6057 262459 262907 52866	80	52866
271253 40850 52866	80	52866
37468 28220 1031 40850 52866	60	52866
28158 271253 40850 52866	60	52866
264409 52866	60	52866

Observation: All surviving routes terminate at AS 52866. The broader ROV deployment effectively neutralized AS 8309's hijack by blocking its routes at the transit tier.

10. Session Summary

Phase	Action	Key Outcome
Setup	Loaded AS topology + Invalid-SET, ran init	76,802 ASes, 495,579 links ready
Announcements	8 ASes announced diverse prefixes	1,366,320 BGP updates, 76,228 affected ASes
Policy	Raised local-pref for 29.214.0.0/16 at AS 34549 via 3356	Route via AS 3356 promoted to localPref 120
Hijack	AS 52866 and AS 8309 announced 187.156.112.139/23	Both hijackers propagated widely across the Internet
ROV (manual)	Enabled ROV at 4 ASes	Hijacker routes partially suppressed; 71,589 affected
ROV (high-deg)	Loaded 7,680-AS ROV set, re-init	AS 8309 fully neutralized; AS 52866 survived as sole origin